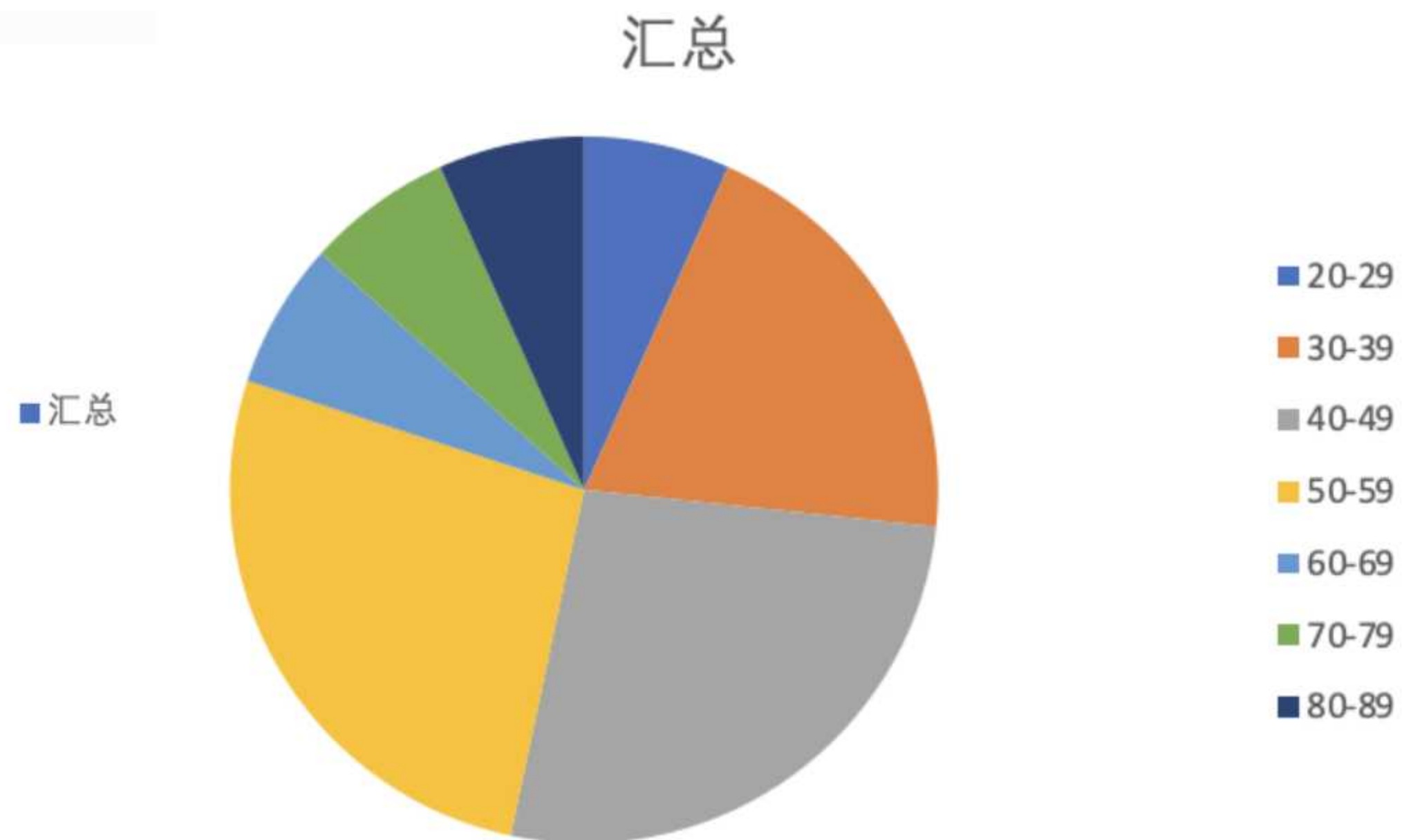
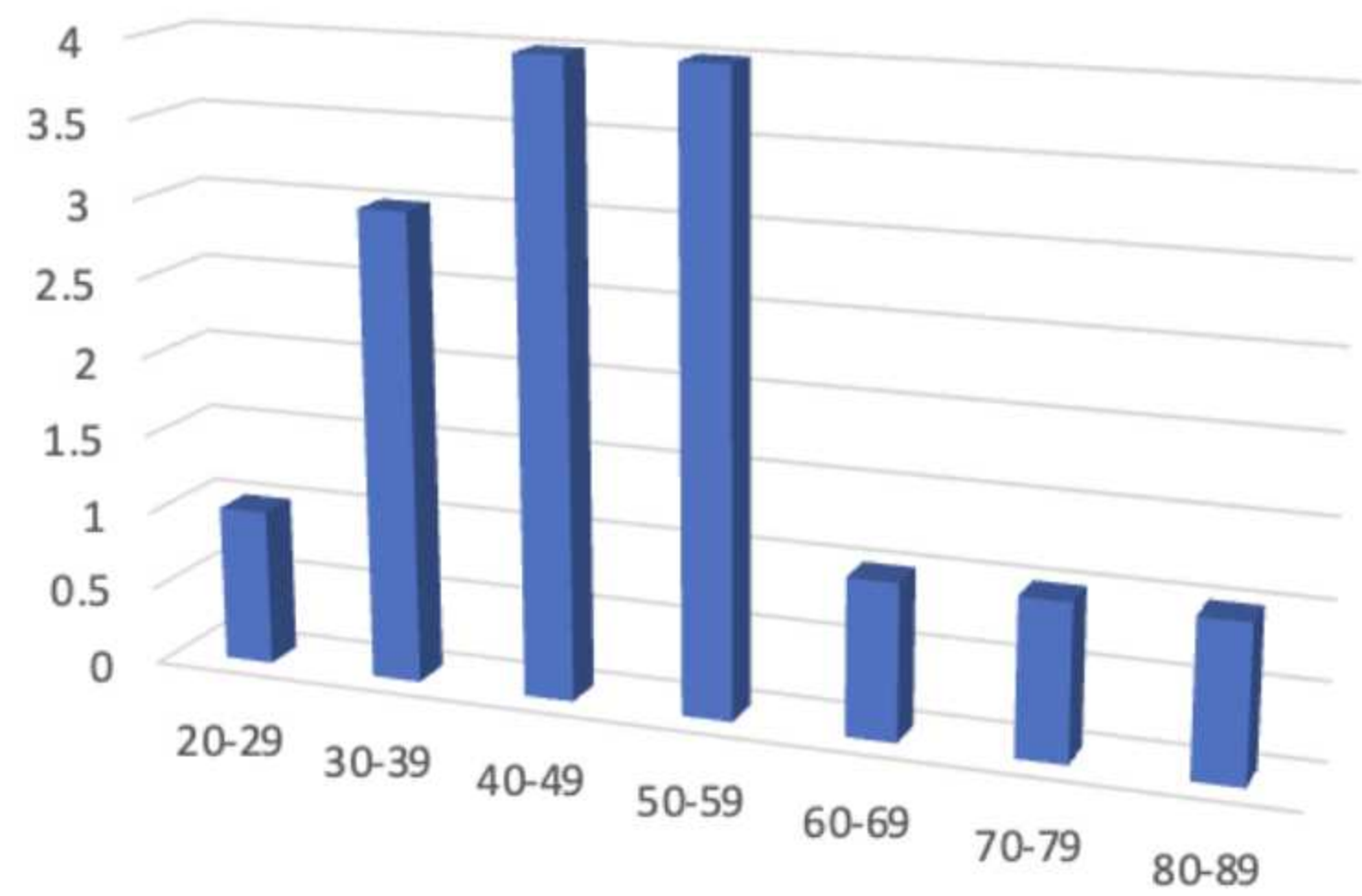


# 机器学习第一次随堂测试

浙江大学

赵洲

# 成绩分布



# 判断题

- ( ) 1. If the data are linear separable, a *Perceptron* can classify every data point in the training data correctly.

# 判断题

- ( T ) 1. If the data are linear separable, a *Perceptron* can classify every data point in the training data correctly.
- True.
- The Perceptron algorithm can find a hyperplane that separates the data perfectly when the data is linearly separable. This is guaranteed by the Perceptron Convergence Theorem.



# 判断题

- () 2. The maximum margin decision boundaries that support vector machines construct have the lowest generalization error among all linear classifiers.

# 判断题

- ( F ) 2. The maximum margin decision boundaries that support vector machines construct have the lowest generalization error among all linear classifiers.
- False.
- SVMs aim to maximize the margin, which often improves generalization, but they do not guarantee the lowest generalization error. Generalization error depends on factors like data distribution, model assumptions, and noise.

# 判断题

- ( ) 3. The standard *K-Means* algorithm is NOT guaranteed to converge.

# 判断题

- ( F ) 3. The standard *K-Means* algorithm is NOT guaranteed to converge.

False.

The standard K-Means algorithm is guaranteed to converge, but it may converge to a local minimum rather than the global minimum. The objective function decreases monotonically and stabilizes.



# 判断题

- ( ) 4. One disadvantage of Q-learning is that it can only be used when the learner has prior knowledge of how its actions affect its environment.

# 判断题

- ( F ) 4. One disadvantage of Q-learning is that it can only be used when the learner has prior knowledge of how its actions affect its environment.
- False.
- Q-learning is a model-free reinforcement learning algorithm. It does not require prior knowledge of the environment's dynamics (transition probabilities). It learns through exploration and experience.

# 判断题

- ( ) 5. There is no exploration-exploitation trade-off in policy iteration and value iteration.

# 判断题

- ( T ) 5. There is no exploration-exploitation trade-off in policy iteration and value iteration.
- True.
- Policy iteration and value iteration are dynamic programming methods that assume full knowledge of the environment's dynamics. They do not involve exploration-exploitation because they compute exact solutions under the assumption of perfect knowledge.



# 判断题

- ( ) 6. For spectral clustering, different initializations will lead to different results.

# 判断题

- ( F ) 6. For spectral clustering, different initializations will lead to different results.
- False.
- Spectral clustering relies on eigenvalue decomposition of the similarity graph's Laplacian matrix. The eigenvalue computation is deterministic. While K-means may still be applied to the spectral embedding, the spectral clustering itself does not depend on initialization.

# 判断题

- ( ) 7. Gradient descent has the problem of sometimes falling into local minima, whereas EM does not.

# 判断题

- ( F ) 7. Gradient descent has the problem of sometimes falling into local minima, whereas EM does not.
- False.
- Both gradient descent and the EM (Expectation-Maximization) algorithm can fall into local minima. EM maximizes the likelihood iteratively but does not guarantee convergence to a global optimum.



# 多选题

- 1) In real world, some problems can be better addressed with Supervised Learning, but some other problems are suitable for Unsupervised Learning. In your opinion, which of the following problems you will apply Unsupervised Learning to? ( )
- A. Project multiple high-dimension vector into 2-D space base on pair-wise Euclidean distance among them.
  - B. Given 50 articles written by male authors, and 50 articles written by female authors, learn to predict the gender of a new manuscript's author (when the identity of this author is unknown).
  - C. Given a large set of essays without topics, group them into different topics according to some predefined similarity.
  - D. Predict the number of registrations (挂号) of Zhejiang University Hospital for a specific day

# 多选题

- 1) In real world, some problems can be better addressed with Supervised Learning, but some other problems are suitable for Unsupervised Learning. In your opinion, which of the following problems you will apply Unsupervised Learning to? ( **AC** )
- A. Project multiple high-dimension vector into 2-D space base on pair-wise Euclidean distance among them.
  - B. Given 50 articles written by male authors, and 50 articles written by female authors, learn to predict the gender of a new manuscript's author (when the identity of this author is unknown).
  - C. Given a large set of essays without topics, group them into different topics according to some predefined similarity.
  - D. Predict the number of registrations (挂号) of Zhejiang University Hospital for a specific day

# 多选题

- **A. Project multiple high-dimension vector into 2-D space based on pair-wise Euclidean distance among them.**
- **• Correct (True):**
- This problem describes dimensionality reduction, which is typically solved using unsupervised learning methods like Principal Component Analysis (PCA), t-SNE, or UMAP. These methods do not require labeled data but work purely based on relationships (e.g., pairwise distances) between data points.
  
- **B. Given 50 articles written by male authors, and 50 articles written by female authors, learn to predict the gender of a new manuscript's author (when the identity of this author is unknown).**
- **• Incorrect (False):**
- This is a Supervised Learning problem because the data comes with labeled categories (male/female authors). The model needs to learn from these labeled examples and predict the gender of a new manuscript. Unsupervised Learning is not appropriate here.

# 多选题

- **C. Given a large set of essays without topics, group them into different topics according to some predefined similarity.**
- **• Correct (True):**
- This is a clustering problem, which falls under unsupervised learning. Methods like K-means, hierarchical clustering, or Latent Dirichlet Allocation (LDA) can be used to group data into topics based on similarities without prior topic labels.
  
- **D. Predict the number of registrations (挂号) of Zhejiang University Hospital for a specific day.**
- **• Incorrect (False):**
- This is a Supervised Learning (regression) problem because it involves predicting a numerical value based on historical labeled data (e.g., past registration numbers). Supervised models like linear regression or neural networks would be appropriate here.



# 多选题

3) For SVM trained on a 2-classes linear separable dataset with  $N$  samples ( $N \geq 4$ ), how many support vectors could there be? ( )

- A. 1
- B. 2
- C.  $\lfloor N/2 \rfloor$  (largest possible integer value which is less than or equal  $N/2$ )
- D. 1.5

# 多选题

3) For SVM trained on a 2-classes linear separable dataset with  $N$  samples ( $N \geq 4$ ), how many support vectors could there be? ( *BC* )

- A. 1
- B. 2
- C.  $\lfloor N/2 \rfloor$  (largest possible integer value which is less than or equal  $N/2$ )
- D. 1.5

# 多选题

- **3) Choice A: 1**

- **• Incorrect.**

- In an SVM for linearly separable data, support vectors **lie on the margin boundaries**. For a 2-class problem in a linearly separable dataset, at least **two support vectors** are required—one for each margin. A single support vector is insufficient to define two distinct margin boundaries.

- **Choice B: 2**

- **• Correct.**

- In the simplest case (e.g., perfectly balanced linearly separable data in 2D), the minimum number of support vectors required is **2**—one from each class that defines the margin. However, the actual number of support vectors could be higher depending on the dataset.

# 多选题

- 3)
- **Choice C:  $\lfloor N/2 \rfloor$  (largest possible integer value less than or equal to  $N/2$ )**
- • **Correct.**
- While SVM aims to find the optimal margin, **multiple samples** can lie on the margin boundaries, making them support vectors. In certain cases, **up to  $\lfloor N/2 \rfloor$**  samples can be support vectors. This happens when the data distribution causes a significant portion of the samples to lie on the margins.
- **Choice D: 1.5**
- • **Incorrect.**
- The number of support vectors must be an **integer**, as they correspond to actual samples in the dataset. “1.5” is not a valid count for support vectors.



# 多选题

4) Which of the following statements are true? ( )

- A. GMM algorithm is guaranteed to converge the global optimum.
- B. Since k-means is an unsupervised learning algorithm, it cannot overfit the data, and thus it is always better to have as large a number of clusters as it is computationally feasible.
- C. For some datasets, the "right" or "correct" value of  $k$  (the number of clusters in k-means) can be ambiguous and hard even for a human expert looking carefully at the data to decide.
- D. Both GMM and k-means are solved by EM algorithm and they will get the same clustering result except GMM can also show that how probable a sample belongs to a cluster.

# 多选题

4) Which of the following statements are true? ( C )

- A. GMM algorithm is guaranteed to converge the global optimum.
- B. Since k-means is an unsupervised learning algorithm, it cannot overfit the data, and thus it is always better to have as large a number of clusters as it is computationally feasible.
- C. For some datasets, the "right" or "correct" value of k (the number of clusters in k-means) can be ambiguous and hard even for a human expert looking carefully at the data to decide.
- D. Both GMM and k-means are solved by EM algorithm and they will get the same clustering result except GMM can also show that how probable a sample belongs to a cluster.

# 多选题

**A. GMM algorithm is guaranteed to converge to the global optimum.**

•False.

The Gaussian Mixture Model (GMM) algorithm, typically solved using the Expectation-Maximization (EM) algorithm, is not guaranteed to converge to the global optimum. EM only guarantees convergence to a local optimum of the likelihood function, which depends on initialization.

**B. Since k-means is an unsupervised learning algorithm, it cannot overfit the data, and thus it is always better to have as large a number of clusters as it is computationally feasible.**

•False.

K-means can overfit the data when the number of clusters  $k$  is too large. Increasing  $k$  allows the algorithm to fit the data points more closely, potentially capturing noise instead of meaningful structure. Therefore, larger  $k$  is not always better.



# 多选题

**C. For some datasets, the “right” or “correct” value of  $k$  (the number of clusters in  $k$ -means) can be ambiguous and hard even for a human expert looking carefully at the data to decide.**

•True.

The “correct” number of clusters  $k$  in  $k$ -means is often ambiguous, particularly in datasets with complex or overlapping structures. Even human experts can find it difficult to determine the appropriate value of  $k$ , as clustering often lacks a single, objectively correct answer.

**D. Both GMM and  $k$ -means are solved by EM algorithm and they will get the same clustering result except GMM can also show how probable a sample belongs to a cluster.**

•False.

While GMM uses the EM algorithm,  $k$ -means does not explicitly use EM—it uses an iterative approach of assigning points to clusters and updating centroids. Additionally, GMM and  $k$ -means do not necessarily produce the same clustering results, as GMM accounts for cluster probabilities and assumes Gaussian distributions, whereas  $k$ -means uses hard assignments based on distance.



# 多选题

- 5) Which of the following statements for Neural Network is correct? ()
- A. Multilayer Neural Network can be viewed as a linear discriminant with a nonlinear transformation for inputs.
  - B. Not all continuous function can be implemented in a 3-layer fully connected Neural Network with sufficient number of hidden units(neurons),
  - C. The training results of Neural Network will vary with different initializations.
  - D. Momentum and Weight Decay are two regularization methods for Neural Network.

# 多选题

- 5) Which of the following statements for Neural Network is correct? ( *AC* )
- A. Multilayer Neural Network can be viewed as a linear discriminant with a nonlinear transformation for inputs.
  - B. Not all continuous function can be implemented in a 3-layer fully connected Neural Network with sufficient number of hidden units(neurons),
  - C. The training results of Neural Network will vary with different initializations.
  - D. Momentum and Weight Decay are two regularization methods for Neural Network.



# 多选题

5) A. Multilayer Neural Network can be viewed as a linear discriminant with a nonlinear transformation for inputs.

- True.

A multilayer neural network applies nonlinear activation functions (like ReLU, Sigmoid, or Tanh) to transform the input data, and the final layer often implements a linear discriminant. Thus, the overall network can be viewed as a linear classifier applied to a nonlinearly transformed input space.

B. Not all continuous functions can be implemented in a 3-layer fully connected Neural Network with sufficient number of hidden units (neurons).

- False.

According to the Universal Approximation Theorem, a fully connected neural network with at least one hidden layer (3 layers total: input, hidden, and output) and a sufficient number of neurons can approximate any continuous function on a compact domain.



# 多选题

5) C. The training results of Neural Network will vary with different initializations.

- True.

Neural networks are typically trained using gradient-based optimization methods like stochastic gradient descent. The network's weights are initialized randomly, which can lead to different local minima, and thus the training results will vary with different initializations.

D. Momentum and Weight Decay are two regularization methods for Neural Network.

- False.

- Momentum is an optimization technique that helps smooth and accelerate gradient descent by adding a fraction of the previous update to the current one. It is not a regularization method.

- Weight Decay (L2 regularization) is a regularization method that penalizes large weights to prevent overfitting.



# 多选题

6) Which of the following classifiers can achieve zero training error on the dataset below? ()

- A. Decision Tree (ID3)
- B. Perceptron
- C. Logistic Regression
- D. Neural Network

| Feature 1 | Feature 2 | Label |
|-----------|-----------|-------|
| X         | X         | F     |
| X         | O         | T     |
| O         | X         | T     |
| O         | O         | F     |

# 多选题

6) Which of the following classifiers can achieve zero training error on the dataset below? ( *AD* )

- A. Decision Tree (ID3)
- B. Perceptron
- C. Logistic Regression
- D. Neural Network

| Feature 1 | Feature 2 | Label |
|-----------|-----------|-------|
| X         | X         | F     |
| X         | O         | T     |
| O         | X         | T     |
| O         | O         | F     |



# 多选题

## 6) A. Decision Tree (ID3)

- True.

A Decision Tree (ID3) can achieve zero training error on this dataset. It can create splits based on Feature 1 and Feature 2 to perfectly separate the labels (F and T). For example:

- If Feature 1 = X and Feature 2 = X  $\rightarrow$  F.
- If Feature 1 = X and Feature 2 = O  $\rightarrow$  T.
- If Feature 1 = O and Feature 2 = X  $\rightarrow$  T.
- If Feature 1 = O and Feature 2 = O  $\rightarrow$  F.

Decision Trees are highly flexible and can fit the training data exactly.

## B. Perceptron

- False.

A Perceptron can only learn linearly separable datasets. In this case, the dataset is not linearly separable in the 2D feature space because the “T” and “F” labels cannot be perfectly divided by a single straight line.

# 多选题

## 6) C. Logistic Regression

- False.

Logistic Regression is also a linear classifier, and it cannot perfectly classify this dataset because it is not linearly separable. The “T” and “F” labels require a more complex decision boundary.

## D. Neural Network

- True.

A Neural Network with at least one hidden layer and a nonlinear activation function can model complex decision boundaries, including those required to separate this dataset. Thus, a neural network can achieve zero training error by fitting the data perfectly.

# 多选题

- 7) Which of the following statements for different loss is correct? ( )
- A. In AdaBoost weights of the misclassified examples go up by the same multiplicative factor.
  - B. In AdaBoost, weighted training error  $\epsilon_t$  of the  $t^{\text{th}}$  weak classifier on training data with weights  $D_t$  tends to increase as a function of  $t$ .
  - C. AdaBoost is actually optimizing the Exponential loss.
  - D. AdaBoost will eventually give zero training error regardless of the type of weak classifier it uses, provided enough iterations are performed.

# 多选题

- 7) Which of the following statements for different loss is correct? ( *ABC* )
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  - C. AdaBoost is actually optimizing the Exponential loss.
  - D. AdaBoost will eventually give zero training error regardless of the type of weak classifier it uses, provided enough iterations are performed.



# 多选题

A. In AdaBoost, weights of the misclassified examples go up by the same multiplicative factor.

•True.

In AdaBoost, the weights of misclassified examples increase by a constant multiplicative factor based on the performance of the weak learner. Specifically, the factor depends on the weak classifier's error rate. Misclassified examples receive higher weights to focus the model on harder examples in subsequent rounds.

B. In AdaBoost, weighted training error  $\epsilon_t$  of the  $t$ -th weak classifier on training data with weights tends to increase as a function of  $t$ .

•False.

The weighted training error  $\epsilon_t$  of the weak classifier does not necessarily increase as a function of  $t$ . AdaBoost aims to reduce this error by focusing on misclassified examples. Although individual weak classifiers may struggle with certain data, the weighted error typically fluctuates rather than strictly increasing.



# 多选题

**C. AdaBoost is actually optimizing the Exponential loss.**

•True.

AdaBoost minimizes the Exponential Loss function. The loss is defined as:

$$L_{\text{exp}} = \sum_{i=1}^N \exp(-y_i \cdot f(x_i)) \quad (1)$$

where  $y_i$  is the true label, and  $f(x_i)$  is the model prediction. The exponential loss strongly penalizes misclassified points, which aligns with how AdaBoost focuses on hard-to-classify examples.

**D. AdaBoost will eventually give zero training error regardless of the type of weak classifier it uses, provided enough iterations are performed.**

•False.

AdaBoost can achieve zero training error only if the weak classifiers are sufficiently expressive to model the dataset. If the weak classifiers are too limited (e.g., very simple or underpowered), AdaBoost cannot perfectly fit the training data, even with many iterations.

# 多选题

- 8) Which of the following statements are true for K-NN classifiers? ( )
- A. The classification accuracy is always better with larger values of K.
  - B. The decision boundary can be non-linear.
  - C. Like other supervised algorithms, K-NN requires an explicit training step.
  - D. The decision boundary is smoother with smaller K.



# 多选题

- 8) Which of the following statements are true for K-NN classifiers? ( **B** )
- A. The classification accuracy is always better with larger values of K.
  - B. The decision boundary can be non-linear.
  - C. Like other supervised algorithms, K-NN requires an explicit training step.
  - D. The decision boundary is smoother with smaller K.

# 多选题

**A. The classification accuracy is always better with larger values of  $K$ .**

- False.

Increasing  $K$  can smooth the decision boundary and reduce overfitting, but it does not always improve accuracy. If  $K$  becomes too large, the model can underfit because it starts ignoring local structure and may misclassify points due to majority voting over distant neighbors.

**B. The decision boundary can be non-linear.**

- True.

$K$ -NN classifiers can produce non-linear decision boundaries because the classification decision depends on the distance to neighboring data points, which does not require any assumption of linearity. The boundaries adapt to the local distribution of the data.



# 多选题

**C. Like other supervised algorithms, K-NN requires an explicit training step.**

•False.

K-NN does not involve an explicit training phase. It is a lazy learning algorithm, meaning it simply stores the training data and classifies new points based on distances to the stored data. No model parameters are learned in advance.

**D. The decision boundary is smoother with smaller K.**

•False.

With smaller  $K$ , the decision boundary becomes more complex and jagged because it is heavily influenced by individual points and noise in the dataset. Larger  $K$  values smooth the decision boundary by averaging over more neighbors.

# 简答题

1. What is the purpose of regularization? For linear regression, what's the difference between L1 and L2 regularization?

# 简答题

1. What is the purpose of regularization? For linear regression, what's the difference between L1 and L2 regularization?

The purpose of regularization is to reduce over-fitting. [2 points]

The differences are: [3 points]

1) The L1 regularization uses L1-norm as regularization term and will results in sparser model parameters.

2) The L2 regularization uses L2-norm as regularization term and will results in smaller model parameters.

# 简答题

2. Describe the difference between a maximum likelihood estimation and a maximum a posteriori estimation. And try to use them to explain Ridge Regression and Linear Regression.



# 简答题

2. Describe the difference between a maximum likelihood estimation and a maximum a posteriori estimation. And try to use them to explain Ridge Regression and Linear Regression.

**Answer:**

- a) MLE finds parameters to maximize the likelihood function, while MAP maximizes the posterior probability. [2 points]
- b) Ridge regression uses the prior information. [3 points]

【其他合理回答亦可】

# 简答题

3. For clustering algorithms, please name two limitations of K-Means Clustering and two limitations of Spectral Clustering.

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3. For clustering algorithms, please name two limitations of K-Means Clustering and two limitations of Spectral Clustering.



■ **【下述仅为示例答案，具体答案合情合理即可得分】**

1) *Limitations of K-Means Clustering [3 points]*

- a) *Results vary with different initializations.*
- b) *Not suitable for non-spherical clusters*

2) *Limitations of Spectral Clustering [2 points]*

- a) *Constructing a adjacency matrix for large datasets is time-consuming*
- b) *Extremely difficult to do eigen decomposition for a very large Laplacian matrix (e.g.  $N = 1,000,000$ )*

# 简答题

4. In this course, we have learned many linear classifiers and nonlinear classifiers. Please compare Logistic Regression (linear) and Decision Tree (non-linear) and list their advantages over the other one. Why we prefer Logistic Regression in the click-through rate (CTR) task of video recommendation?
  - CTR: estimate the probability of a user clicking on the item



# 简答题

4. In this course, we have learned many linear classifiers and nonlinear classifiers. Please compare Logistic Regression (linear) and Decision Tree (non-linear) and list their advantages over the other one. Why we prefer Logistic Regression in the click-through rate (CTR) task of video recommendation?
- CTR: estimate the probability of a user clicking on the item
  - 
  - 
  - Answer:
    - a) Logistic Regression: require much less computation; online learning; stable performance with extremely large feature space; low variance 【一点2分】
    - b) Random Forest: high capacity when feature space is small; nonlinear, low bias; interpretable rule; 【一点即给2分】
    - c) Online learning 【1分】

# 计算题

## ■ 4. Naïve Bayes Classification (12 points)

- Consider the following dataset with two classes. The features in the dataset is independent, and we guarantee all possible values of each feature are presented.

a) Estimate the conditional probabilities for  $P(A|+)$ ,  $P(B|+)$ ,  $P(C|+)$ ,  $P(A|-)$ ,  $P(B|-)$ , and  $P(C|-)$ .

| Record | A | B | C | Class |
|--------|---|---|---|-------|
| 1      | 0 | 0 | 1 | -     |
| 2      | 1 | 0 | 1 | +     |
| 3      | 0 | 1 | 0 | -     |
| 4      | 1 | 0 | 0 | -     |
| 5      | 1 | 0 | 1 | +     |
| 6      | 0 | 0 | 1 | +     |
| 7      | 1 | 1 | 0 | -     |
| 8      | 0 | 0 | 0 | -     |
| 9      | 0 | 1 | 0 | +     |
| 10     | 1 | 1 | 1 | +     |

# 计算题

## ■ 4. Naïve Bayes Classification (12 points)

- Consider the following dataset with two classes. The features in the dataset is independent, and we guarantee all possible values of each feature are presented.

a)  $P(A=1|+)=3/5$ ,  $P(A=0|+)=2/5$ ,  $P(A=1|-)=2/5$ ,  
 $P(A=0|-)=3/5$  [2 points]  
 $P(B=1|+)=2/5$ ,  $P(B=0|+)=3/5$ ,  $P(B=1|-)=2/5$ ,  
 $P(B=0|-)=3/5$  [2 points]  
 $P(C=1|+)=4/5$ ,  $P(C=0|+)=1/5$ ,  $P(C=1|-)=1/5$ ,  
 $P(C=0|-)=4/5$  [2 points]

| Record | A | B | C | Class |
|--------|---|---|---|-------|
| 1      | 0 | 0 | 1 | -     |
| 2      | 1 | 0 | 1 | +     |
| 3      | 0 | 1 | 0 | -     |
| 4      | 1 | 0 | 0 | -     |
| 5      | 1 | 0 | 1 | +     |
| 6      | 0 | 0 | 1 | +     |
| 7      | 1 | 1 | 0 | -     |
| 8      | 0 | 0 | 0 | -     |
| 9      | 0 | 1 | 0 | +     |
| 10     | 1 | 1 | 1 | +     |

# 计算题

## ■ 4. Naïve Bayes Classification (12 points)

- Consider the following dataset with two classes. The features in the dataset is independent, and we guarantee all possible values of each feature are presented.

b) Use the estimate of conditional probabilities in the previous question to predict the class label for the following test samples using the Naive Bayes model.:

*i.  $(A = 1, B = 1, C = 1)$*

*ii.  $(A = 0, B = 0, C = 0)$*

| Record | A | B | C | Class |
|--------|---|---|---|-------|
| 1      | 0 | 0 | 1 | -     |
| 2      | 1 | 0 | 1 | +     |
| 3      | 0 | 1 | 0 | -     |
| 4      | 1 | 0 | 0 | -     |
| 5      | 1 | 0 | 1 | +     |
| 6      | 0 | 0 | 1 | +     |
| 7      | 1 | 1 | 0 | -     |
| 8      | 0 | 0 | 0 | -     |
| 9      | 0 | 1 | 0 | +     |
| 10     | 1 | 1 | 1 | +     |



# 计算题

## ■ 4. Naïve Bayes Classification (12 points)

- Consider the following dataset with two classes. The features in the dataset is independent, and we guarantee all possible values of each feature are presented.

b) 【每问1分】

- $P(A=1, B=1, C=1, +)=12/125$ ,  $P(A=1, B=1, C=1, -)=2/125$ , label is +
- $P(A=0, B=0, C=0, +)=3/125$ ,  $P(A=0, B=0, C=0, -)=18/125$ , label is -

| Record | A | B | C | Class |
|--------|---|---|---|-------|
| 1      | 0 | 0 | 1 | -     |
| 2      | 1 | 0 | 1 | +     |
| 3      | 0 | 1 | 0 | -     |
| 4      | 1 | 0 | 0 | -     |
| 5      | 1 | 0 | 1 | +     |
| 6      | 0 | 0 | 1 | +     |
| 7      | 1 | 1 | 0 | -     |
| 8      | 0 | 0 | 0 | -     |
| 9      | 0 | 1 | 0 | +     |
| 10     | 1 | 1 | 1 | +     |

# 计算题

## ■ 4. Naïve Bayes Classification (12 points)

- Consider the following dataset with two classes. The features in the dataset is independent, and we guarantee all possible values of each feature are presented.

c) Please explain why the naïve Bayesian model is called naïve.

| Record | A | B | C | Class |
|--------|---|---|---|-------|
| 1      | 0 | 0 | 1 | -     |
| 2      | 1 | 0 | 1 | +     |
| 3      | 0 | 1 | 0 | -     |
| 4      | 1 | 0 | 0 | -     |
| 5      | 1 | 0 | 1 | +     |
| 6      | 0 | 0 | 1 | +     |
| 7      | 1 | 1 | 0 | -     |
| 8      | 0 | 0 | 0 | -     |
| 9      | 0 | 1 | 0 | +     |
| 10     | 1 | 1 | 1 | +     |

# 计算题

## ■ 4. Naïve Bayes Classification (12 points)

- Consider the following dataset with two classes. The features in the dataset is independent, and we guarantee all possible values of each feature are presented.

c) Because it assumes the features of the data are independent. [2 points]

| Record | A | B | C | Class |
|--------|---|---|---|-------|
| 1      | 0 | 0 | 1 | -     |
| 2      | 1 | 0 | 1 | +     |
| 3      | 0 | 1 | 0 | -     |
| 4      | 1 | 0 | 0 | -     |
| 5      | 1 | 0 | 1 | +     |
| 6      | 0 | 0 | 1 | +     |
| 7      | 1 | 1 | 0 | -     |
| 8      | 0 | 0 | 0 | -     |
| 9      | 0 | 1 | 0 | +     |
| 10     | 1 | 1 | 1 | +     |

# 计算题

## ■ 4. Naïve Bayes Classification (12 points)

- Consider the following dataset with two classes. The features in the dataset is independent, and we guarantee all possible values of each feature are presented.

d) Suppose you know for fact that A, B, C are independent random variables. In this case is it possible for any other classifier (e.g., a decision tree or a neural net) to do better than a naive Bayes classifier? And why? (The dataset is irrelevant for this question).

| Record | A | B | C | Class |
|--------|---|---|---|-------|
| 1      | 0 | 0 | 1 | -     |
| 2      | 1 | 0 | 1 | +     |
| 3      | 0 | 1 | 0 | -     |
| 4      | 1 | 0 | 0 | -     |
| 5      | 1 | 0 | 1 | +     |
| 6      | 0 | 0 | 1 | +     |
| 7      | 1 | 1 | 0 | -     |
| 8      | 0 | 0 | 0 | -     |
| 9      | 0 | 1 | 0 | +     |
| 10     | 1 | 1 | 1 | +     |



# 计算题

## ■ 4. Naïve Bayes Classification (12 points)

- Consider the following dataset with two classes. The features in the dataset is independent, and we guarantee all possible values of each feature are presented.

d) Yes. The independency of A, B, C does not imply that they are independent within each class (in other words, they are not necessarily independent when conditioned on  $y$ ). Therefore, naive Bayes classifier may not be able to model the function well, while a decision tree might. Thus, for example,  $y = A \text{ XOR } B$ , is an example where A, B might be independent variables, but a naive Bayes classifier will not model the function well since for a particular class (say,  $y = 0$ ), A and B are dependent.

| Record | A | B | C | Class |
|--------|---|---|---|-------|
| 1      | 0 | 0 | 1 | -     |
| 2      | 1 | 0 | 1 | +     |
| 3      | 0 | 1 | 0 | -     |
| 4      | 1 | 0 | 0 | -     |
| 5      | 1 | 0 | 1 | +     |
| 6      | 0 | 0 | 1 | +     |
| 7      | 1 | 1 | 0 | -     |
| 8      | 0 | 0 | 0 | -     |
| 9      | 0 | 1 | 0 | +     |
| 10     | 1 | 1 | 1 | +     |

# 计算题

## ■ 5. PCA & SVD (10 points)

- a) Consider the standard least-squares linear regression problem ( $w^T X \approx y^T$ ), where the unknown parameters to be learned are the column vector  $w$ . As we learned from linear regression, the solution is  $w = (X^T X)^{-1} X^T y$ . Let the singular value decomposition (SVD) of  $X$  be  $X = USV^T$ . Show that the solution can be simplified to  $w = VS^{-1}U^T y$ .



# 计算题

## ■ 5. PCA & SVD (10 points)

- a) Consider the standard least-squares linear regression problem ( $\mathbf{w}^T \mathbf{X} \approx \mathbf{y}^T$ ), where the unknown parameters to be learned are the column vector  $\mathbf{w}$ . As we learned from linear regression, the solution is  $\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ . Let the singular value decomposition (SVD) of  $\mathbf{X}$  be  $\mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ . Show that the solution can be simplified to  $\mathbf{w} = \mathbf{V} \mathbf{S}^{-1} \mathbf{U}^T \mathbf{y}$ .

■ *Answer:*

$$\because \mathbf{U}^T \mathbf{U} = \mathbf{V}^T \mathbf{V} = \mathbf{I}, \text{ and } \mathbf{X} = \mathbf{U} \mathbf{S} \mathbf{V}^T$$

a) 
$$\therefore \mathbf{w} = (\mathbf{V} \mathbf{S}^T \mathbf{S} \mathbf{V}^T)^{-1} \mathbf{V} \mathbf{S}^T \mathbf{U}^T \mathbf{y} = \mathbf{V} \mathbf{S}^{-1} \mathbf{S}^{-T} \mathbf{V}^T \mathbf{V} \mathbf{S}^T \mathbf{U} \mathbf{y} = \mathbf{V} \mathbf{S}^{-1} \mathbf{U}^T \mathbf{y}$$

**【能正确使用答案第一行公式推导得3分，在此基础上其余细节准确得6分满分】**

# 计算题

## ■ 5. PCA & SVD (10 points)

b) What is the relation between SVD and PCA?



# 计算题

## ■ 5. PCA & SVD (10 points)

b) What is the relation between SVD and PCA?

*Answer:*

*b) If the data is centralized (zero mean), PCA can be done through SVD. And the eigenvalues in PCA is the square of the singular values in SVD.*

**【能答出平方关系或二者等价得2分，在此基础上能答出去中心化得4分满分】**

# 计算题

## ■ 5. PCA & SVD (10 points)

- c) Both PCA and linear regression can be thought of as algorithms for minimizing a sum of squared errors. Explain which error is being minimized in each algorithm.

■ *Answer:*

# 计算题

## ■ 5. PCA & SVD (10 points)

■ *Answer:*

- In PCA, the goal is to find a lower-dimensional representation of the data that captures the maximum variance. The error being minimized in PCA is the **reconstruction error**, which is the squared distance between each data point and its projection onto the lower-dimensional subspace.
- Mathematically, if the original data point is  $\mathbf{x}_i$  and its projection onto the subspace is  $\hat{\mathbf{x}}_i$ , the reconstruction error is:

$$\text{Reconstruction Error} = \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2. \quad (1)$$

- Here,  $\hat{\mathbf{x}}_i$  is obtained by projecting  $\mathbf{x}_i$  onto the principal components (the directions of maximum variance). PCA minimizes this error by ensuring that the principal components are the eigenvectors of the covariance matrix of the

# 计算题

- **5. PCA & SVD (10 points)**

- *Answer:* **Linear Regression**

- In linear regression, the goal is to fit a line (or hyperplane) to predict a target variable from the input features . The error being minimized in linear regression is the **prediction error** or **residual sum of squares (RSS)**, which is the squared difference between the true target values and the predicted values.
- Mathematically, if  $y_i$  is the actual target value and  $\hat{y}_i = w^T x_i + b$  is the predicted value, the error is:

$$\text{Prediction Error} = \sum_{i=1}^N (y_i - \hat{y}_i)^2. \quad (2)$$

- Here, the line (or hyperplane) is chosen such that the sum of squared residuals (errors between the predictions and the actual values) is minimized.



# 计算题

## ■ 5. PCA & SVD (10 points)

■ *Answer:*

### ■ Summary

- • **PCA** minimizes the **reconstruction error**: the squared distance between data points and their projections onto a lower-dimensional subspace.
- • **Linear Regression** minimizes the **prediction error**: the squared difference between the actual target values and the predicted values.

# 计算题

## ■ 6. Decision Tree (12 points)

- Consider the following dataset for predicting the risk of a loan application based on the applicant's credit history, debt and income.

Which one of those three predicting attributes (Credit history, Debt, or Income) would be selected as the root of a *Decision Tree* using *ID3* algorithm?

The logarithm for selected values are provided:

|    | Credit History | Debt        | Income        | Risk            |
|----|----------------|-------------|---------------|-----------------|
| 1  | <i>Bad</i>     | <i>Low</i>  | <i>0-15</i>   | <i>High</i>     |
| 2  | <i>Bad</i>     | <i>High</i> | <i>15-35</i>  | <i>High</i>     |
| 3  | <i>Bad</i>     | <i>Low</i>  | <i>0-15</i>   | <i>High</i>     |
| 4  | <i>Unknown</i> | <i>High</i> | <i>15-35</i>  | <i>High</i>     |
| 5  | <i>Unknown</i> | <i>High</i> | <i>0-15</i>   | <i>High</i>     |
| 6  | <i>Good</i>    | <i>High</i> | <i>0-15</i>   | <i>High</i>     |
| 7  | <i>Bad</i>     | <i>Low</i>  | <i>&gt;35</i> | <i>Moderate</i> |
| 8  | <i>Unknown</i> | <i>Low</i>  | <i>15-35</i>  | <i>Moderate</i> |
| 9  | <i>Good</i>    | <i>High</i> | <i>15-35</i>  | <i>Moderate</i> |
| 10 | <i>Unknown</i> | <i>Low</i>  | <i>&gt;35</i> | <i>Low</i>      |
| 11 | <i>Unknown</i> | <i>Low</i>  | <i>&gt;35</i> | <i>Low</i>      |
| 12 | <i>Good</i>    | <i>Low</i>  | <i>&gt;35</i> | <i>Low</i>      |
| 13 | <i>Good</i>    | <i>High</i> | <i>&gt;35</i> | <i>Low</i>      |
| 14 | <i>Good</i>    | <i>High</i> | <i>&gt;35</i> | <i>Low</i>      |

Show all the key steps of the calculations.

| $x$         | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | $\frac{1}{5}$ | $\frac{2}{5}$ | $\frac{3}{5}$ | $\frac{1}{6}$ | $\frac{5}{6}$ | $\frac{1}{7}$ | $\frac{2}{7}$ | $\frac{3}{7}$ | $\frac{4}{7}$ |
|-------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| $-\log_2 x$ | 1             | 2             | 0.4           | 2.3           | 1.3           | 0.7           | 2.5           | 0.2           | 2.8           | 1.8           | 1.2           | 0.8           |



# 计算题

## ■ 6. Decision Tree (12 points)

- To determine which attribute (Credit History, Debt, or Income) should be selected as the root of the Decision Tree using the ID3 algorithm, we need to calculate the **Information Gain** for each attribute. The attribute with the highest Information Gain will be chosen as the root.

## ■ Step 1: Calculate the Entropy of the Entire Dataset

**Entropy (S)** measures the impurity or uncertainty in the dataset.

- $$\text{Entropy}(S) = - \sum_{i=1}^c p_i \log_2 p_i$$

- Where:

- $p_i$  is the probability of class

- •  $c$  is the number of classes

**Classes:** High, Moderate, Low

**Counts:**

- High: 6
- Moderate: 3
- Low: 5
- **Total:** 14

Entropy(S)

$$= - \left( \frac{6}{14} \log_2 \frac{6}{14} + \frac{3}{14} \log_2 \frac{3}{14} + \frac{5}{14} \log_2 \frac{5}{14} \right)$$

(S)  $\approx$  1.53 bits

# 计算题

- **6. Decision Tree (12 points)**
- **Step 2: Calculate Information Gain for Each Attribute**
- **Information Gain (IG)** measures the reduction in entropy achieved by partitioning the dataset based on an attribute.
- $IG(S, A) = Entropy(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} Entropy(S_v)$
- **2.1. Attribute: Credit History**  
**Possible Values:** Bad, Unknown, Good
- • **Credit History = Bad**
- • **Examples:** 1, 2, 3, 7
- • **Risks:** 3 High, 1 Moderate
- $Entropy = - \left( \frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right) \approx 0.81 \text{ bits}$
- • **Credit History = Unknown**
- • **Examples:** 4, 5, 8, 10, 11
- • **Risks:** 2 High, 1 Moderate, 2 Low       $Entropy \approx 1.52 \text{ bits}$



# 计算题

- **6. Decision Tree (12 points)**
- **Credit History = Good**
- • **Examples:** 6, 9, 12, 13, 14
- • **Risks:** 1 High, 1 Moderate, 3 Low
- • **Entropy**  $\approx 0.97$  bits
- **Weighted Entropy:**  $\frac{4}{14} \times 0.81 + \frac{5}{14} \times 1.52 + \frac{5}{14} \times 0.97 \approx 1.12$  bits
- **Information Gain:**  $IG(S, \text{Credit History}) = 1.53 - 1.12 \approx 0.41$  bits

# 计算题

- **6. Decision Tree (12 points)**
- **2.2. Attribute: Debt**
- **Possible Values: Low, High**
- • **Debt = Low**
- • **Examples: 1, 3, 7, 8, 10, 11, 12**
- • **Risks: 2 High, 2 Moderate, 3 Low**
- **Entropy  $\approx 1.56$  bits**
- **Debt = High**
- • **Examples: 2, 4, 5, 6, 9, 13, 14**
- • **Risks: 4 High, 1 Moderate, 2 Low    Entropy  $\approx 1.38$  bits**

# 计算题

- 6. Decision Tree (12 points)
- Weighted Entropy:  $\frac{7}{14} \times 1.56 + \frac{7}{14} \times 1.38 \approx 1.47$  bits
- Information Gain:  $IG(S, Debt) = 1.53 - 1.47 \approx 0.06$  bits
- 2.3. Attribute: Income
- Possible Values: 0–15, 15–35, >35
- • Income = 0–15
- • Examples: 1, 3, 5, 6
- • Risks: 4 High
- • Entropy = 0 bits

# 计算题

- 6. Decision Tree (12 points)
- **Income = 15-35**
- • **Examples:** 2, 4, 8, 9
- • **Risks:** 2 High, 2 Moderate
- Entropy = 1 bit
- **Income = >35**
- • **Examples:** 7, 10, 11, 12, 13, 14
- • **Risks:** 1 Moderate, 5 Low
- Entropy  $\approx 0.65$  bits



# 计算题

- **6. Decision Tree (12 points)**

- **Weighted Entropy:**  $\frac{4}{14} \times 0 + \frac{4}{14} \times 1 + \frac{6}{14} \times 0.65 \approx 0.56$  bits

- **Information Gain:**  $IG(S, \text{Income}) = 1.53 - 0.56 \approx 0.97$  bits

- **Step 3: Determine the Root Attribute**

- **Comparing the Information Gains:**

- • **Credit History:** 0.41 bits

- • **Debt:** 0.06 bits

- • **Income:** 0.97 bits

- **Conclusion:** The Income attribute has the highest Information Gain and is therefore selected as the root of the Decision Tree.

# 计算题

## ■ 7. Multi-classes Classifier (16 points)

- As we discussed in class, there are three different ways to construct a multi-classes classifier using multiple binary classifiers, now let's have a try on One-One method and One-Rest method.
- Assume we have a large enough dataset  $D$  with  $K$  classes, and we want to construct a classifier for this dataset using multiple binary classifiers using the two methods mentioned above. Now please answer the following two questions for each method:
  - ◆ How many binary classifiers do we need?
  - ◆ Given  $N$  trained binary classifiers  $\{g_1, \dots, g_N\}$  ( $N$  is the number of binary classifiers in the previous question), please write out the pseudocode of the decision procedure.

# 计算题

## ■ 7. Multi-classes Classifier (16 points)

### ■ Answer:(第一问)

#### 1. One-One method:

a)  $N = K(K - 1)/2$  [2 points]

b) *function Classify(x):*

*for each class c:*

*votes[c] = 0*

*for i from 1 to K-1:*

*for j from i+1 to K:*

*if classifier[i][j] classify x as i:*

*votes[i] += 1*

*else:*

*votes[j] += 1*

*label of x = argmax(votes)* 【6分，不要求代码一致，算法思想相同即可】



# 计算题

- **7. Multi-classes Classifier (16 points)**

- *Answer: (第二问)*

- 1. *One-Rest method:*

- a)  $N = K$  [2 points]

- b) *function Classify(x):*

- 

- *for each classifier g[i]:*

- $\text{confidences}[i] = \text{confidence of } x \text{ given by } g[i]$

- $\text{label of } x = \text{argmax}(\text{confidences})$

- **【6分，不要求代码一致，算法思想相同即可，投票未使用置信度扣2分】**



End